

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12401101
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Utsumi; Shintaro et al.

Bipolar battery stack

Abstract

Disclosed is a bipolar battery with which thermal deterioration of the electrode body due to the generation of heat of tabs can be suppressed. The bipolar battery of the present disclosure includes a first member, a second member, and a laminate electrode body arranged therebetween, wherein the laminate electrode body includes a first current collector constituting a lamination direction end surface, a second current collector constituting the other lamination direction end surface, at least one bipolar current collector arranged between the first current collector and the second current collector, and a plurality of power generating elements which are electrically connected in series via the bipolar current collector between the first current collector and the second current collector, the first current collector is arranged between the first member and the bipolar current collector, the second current collector is arranged between the second member and the bipolar current collector, the first current collector has a first tab, the second current collector has a second tab, an amount of heat generated by the first tab during energization of the battery is greater than an amount of heat generated by the second tab, the first member is a cooling member for cooling the first current collector, and a cooling performance of the first member is greater than a cooling performance of the second member.

Inventors:	Utsumi; Shintaro (Susono, JP), Nomoto; Kazushige (Miyoshi, JP)
Applicant:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
Family ID:	1000008782159
Assignee:	TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota, JP)
Appl. No.:	18/407769
Filed:	January 09, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240154276 A1	May. 09, 2024

Foreign Application Priority Data

JP 2020-030364 Feb. 26, 2020

Related U.S. Application Data

continuation parent-doc US 18177187 20230302 PENDING child-doc US 18407769
continuation parent-doc US 17134768 20201228 US 11621463 child-doc US 18177187

Publication Classification

Int. Cl.: H01M10/04 (20060101); H01M10/613 (20140101); H01M10/655 (20140101);
H01M10/6554 (20140101); H01M10/658 (20140101); H01M50/54 (20210101);
H01M50/562 (20210101); H01M50/574 (20210101); H01M50/578 (20210101);
H01M50/581 (20210101); H01M50/583 (20210101)

U.S. Cl.:

CPC H01M50/581 (20210101); H01M10/0418 (20130101); H01M10/044 (20130101);
H01M10/613 (20150401); H01M10/655 (20150401); H01M10/6554 (20150401);
H01M10/658 (20150401); H01M50/54 (20210101); H01M50/562 (20210101);
H01M50/574 (20210101); H01M50/578 (20210101); H01M50/583 (20210101);
H01M2200/10 (20130101)

Field of Classification Search

CPC: H01M (50/581); H01M (50/54); H01M (50/562); H01M (50/574); H01M (50/578);
H01M (50/583); H01M (10/0418); H01M (10/044); H01M (10/613); H01M (10/655);
H01M (10/6554); H01M (10/658); H01M (2200/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2009/0233164	12/2008	Shimamura et al.	N/A	N/A
2018/0053962	12/2017	Okumura	N/A	H01M 10/282
2018/0131035	12/2017	Hojo et al.	N/A	N/A
2018/0205118	12/2017	Chu et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2012248354	12/2011	JP	N/A
2018077977	12/2017	JP	N/A
2019036397	12/2018	JP	N/A
2006062204	12/2005	WO	N/A

OTHER PUBLICATIONS

Shintaro Utsumi et al., U.S. Appl. No. 17/134,768, Restriction Requirement dated Sep. 27, 2022.
cited by applicant

Primary Examiner: Wang; Eugenia

Attorney, Agent or Firm: Dickinson Wright, PLLC

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/177,187, filed Mar. 2, 2023, which is a continuation of U.S. patent application Ser. No. 17/134,768, filed Dec. 28, 2020, issued as U.S. Pat. No. 11,621,463, on Apr. 4, 2023, which claims priority to JP Patent Application No. 2020-030364, filed Feb. 26, 2020, the specification, claims and abstract of which are incorporated herein by reference in their entireties.

FIELD

(1) The present application discloses a bipolar battery and a bipolar battery stack.

BACKGROUND

(2) Patent Literature 1 discloses a bipolar battery comprising a plurality of bipolar electrodes laminated in series and electrolyte layers interposed therebetween, wherein each bipolar electrode comprises a positive electrode formed on one surface of a current collector and a negative electrode formed on the other surface thereof. In Patent Literature 1, outermost current collectors are provided on opposite ends of the plurality of bipolar electrodes in the lamination direction, and the outermost current collectors are connected to the respective tabs (leads).

(3) Patent Literature 2 discloses a battery pack in which cooling members are provided between unit cells. Furthermore, Patent Literature 3 discloses a battery cooling system, wherein an electrode block is arranged between battery electrode tabs, a cooling member is put in contact with the electrode block, and the electrode tabs are cooled via the electrode block.

CITATION LIST

Patent Literature

(4) [PTL 1] WO 2006/062204 [PTL 2] Japanese Unexamined Patent Publication No. 2019-036397 [PTL 2] Japanese Unexamined Patent Publication No. 2012-248354

SUMMARY

Technical Problem

(5) In accordance with the new findings of the present inventors, in bipolar batteries as disclosed in Patent Literature 1, during energization of the battery, the tabs connected to the outermost current collectors generate heat, and the heat may diffuse from the tabs toward the interior of the electrode body via the outermost current collectors. In other words, there is a risk of thermal deterioration of part of the electrode body due to the heat generated by the tabs. Furthermore, in accordance with the new findings of the present inventors, the amount of heat generated during energization of the battery differs between the positive electrode side tab and the negative electrode side tab provided in the bipolar battery. Thus, it is considered that the above thermal deterioration is likely to occur in the vicinity of the outermost current collectors connected to the tabs, which generate significant quantities of heat.

Solution to Problem

(6) As one means for solving the above problem, the present application discloses: a bipolar battery, comprising a first member, a second member, and a laminate electrode body arranged between the first member and the second member, wherein the laminate electrode body comprises a

first current collector constituting a lamination direction end surface, a second current collector constituting the other lamination direction end surface, at least one bipolar current collector arranged between the first current collector and the second current collector, and a plurality of power generating elements which are electrically connected in series via the bipolar current collector between the first current collector and the second current collector, the first current collector is arranged between the first member and the bipolar current collector, the second current collector is arranged between the second member and the bipolar current collector, the first current collector has a first tab, the second current collector has a second tab, an amount of heat generated by the first tab during energization of the battery is greater than an amount of heat generated by the second tab, the first member is a cooling member for cooling the first current collector, and a cooling performance of the first member is greater than a cooling performance of the second member.

(7) In the bipolar battery of the present disclosure, the first tab may have a shutdown mechanism.

(8) In the bipolar battery of the present disclosure, the second member may be a heat insulating member.

(9) As one means for solving the above problem, the present application discloses a bipolar battery stack comprising a plurality of the above bipolar batteries of the present disclosure.

(10) In the bipolar battery stack of the present disclosure, the second member of one of the bipolar batteries may be laminated on the first member of another bipolar battery.

(11) In the bipolar battery stack of the present disclosure, the first member of one of the bipolar batteries may be laminated on the first member of another bipolar battery.

(12) In the bipolar battery stack of the present disclosure, the first member is shared by one of the bipolar batteries and another of the bipolar batteries.

Advantageous Effects of Invention

(13) According to the technology of the present disclosure, the outermost current collector (first current collector) which is connected to the first tab, which generates significant amounts of heat during energization, can be cooled by the first member. In other words, even when heat is generated by the first tab, the heat is unlikely to spread from the first tab toward the interior of the laminate electrode body via the first current collector, whereby thermal deterioration of the laminate electrode body due to the heat generated by the tab can be suppressed.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic view showing an example of the appearance of a bipolar battery.

(2) FIG. 2 is a schematic view showing the structure of a cross-section taken along line II-II of FIG. 1.

(3) FIG. 3 is a schematic view showing an example of the appearance of a bipolar battery stack.

(4) FIG. 4A is a schematic view detailing the laminate structure of the bipolar battery stack shown in FIG. 3. The exterior body has been omitted.

(5) FIG. 4B is a schematic view detailing another example of the laminate structure of the bipolar battery stack. The exterior body has been omitted.

(6) FIG. 4C is a schematic view detailing another example of the laminate structure of the bipolar battery stack. The exterior body has been omitted.

(7) FIG. 5 is a schematic view detailing an example of the structure of a bipolar battery stack comprising a heat exchanger. The exterior body has been omitted.

DESCRIPTION OF EMBODIMENTS

(8) 1. Bipolar Battery

(9) The structure of a bipolar battery **100** is schematically illustrated in FIGS. 1 and 2. The bipolar

battery **100** comprises a first member **10**, a second member **20**, and a laminate electrode body **30** arranged between the first member **10** and the second member **20**. The laminate electrode body **30** comprises a first current collector **31** constituting a lamination direction end surface, a second current collector **32** constituting the other lamination direction end surface, at least one bipolar current collector **33** arranged between the first current collector **31** and the second current collector **32**, and a plurality of power generating elements **34** which are electrically connected in series via the bipolar current collector **33** between the first current collector **31** and the second current collector **32**. The first current collector **31** is arranged between the first member **10** and the bipolar current collector **33**. The second current collector **32** is arranged between the second member **20** and the bipolar current collector **33**. The first current collector **31** has a first tab **31a**. The second current collector **32** has a second tab **32a**. During energization of the battery **100**, the amount of heat generated by the first tab **31a** is greater than an amount of heat generated by the second tab **32a**. The first member **10** is a cooling member for cooling the first current collector **31**. The cooling performance of the first member **10** is greater than the cooling performance of the second member **20**.

(10) 1.1 First Member

(11) The first member **10** is a cooling member for cooling the first current collector **31**. The shape, structure and material of the first member **10** are not particularly limited as long as it is capable of cooling the first current collector **31**. The first member **10** can be, for example, a plate-like member. Specifically, the first member **10** may be a cooling plate which removes heat from the first current collector **31**. Alternatively, the first member **10** may be a cooling plate having passages of coolant. As will be described later, the first member **10** may be connected to a heat exchanger **50** (refer to FIG. 5). For example, when the first member **10** has coolant passages, the coolant passages may be connected to the heat exchanger **50**.

(12) As shown in FIG. 2, an intermediate member such as an exterior body **40** may be present between the first member **10** and the first current collector **31**. Alternatively, the first member **10** may be in direct contact with the surface of the first current collector **31**.

(13) As shown in FIG. 2, the first member **10** may be provided on only one end side (a position facing the surface of the first current collector **31**) of the laminate electrode body **30** in the lamination direction. Alternatively, the first member **10** may have a portion which protrudes from the position facing the surface of the first current collector **31**, and, for example, the first member **10** may be present on the side surface side of the laminate electrode body **30** by bending the protruding portion (refer to FIG. 5).

(14) The cooling performance of the first member **10** is greater than the cooling performance of the second member **20**. The phrase “cooling performance of the first member **10**” means the ability of the first member **10** to cool the first current collector **31**. The phrase “cooling performance of the second member **20**” means the ability of the second member **20** to cool the second current collector **32**. In other words, the phrase “the cooling performance of the first member **10** is greater than the cooling performance of the second member **20**” means that when the thermal conductivity and heat generation amount of the first current collector **31** are the same as the thermal conductivity and heat generation amount of the second current collector **32**, the amount of heat removed from the first current collector **31** to the first member **10** per unit time is greater than the amount of heat removed from the second current collector **32** to the second member **20** per unit time. Note that as will be described later, the second member **20** need not necessarily be a cooling member, but rather may be, for example, an insulating member. When the second member **20** is an insulating member, the cooling performance of the first member **10** will naturally be higher than the cooling performance of the second member **20**.

(15) 1.2 Second Member

(16) The second member **20** may be a cooling member which cools the second current collector **32**, or may be a non-cooling member which is not intended to cool the second current collector **32**. The

shape, structure, and material of the second member **20** are not particularly limited as long as the cooling performance thereof is less than that of the first member **10**. The second member **20** may be a heat insulating member. For example, the second member **10** may be a plate-like insulating member for preventing the spread of fire. Alternatively, the second member **20** may be a resin frame for the retention of the laminate electrode body **30** or the exterior body **40**. Note that in the present application, such a resin frame is also regarded as a type of heat insulating member.

(17) As shown in FIG. 2, an intermediate member such as the exterior body **40** may be present between the second member **20** and the second current collector **32**. Alternatively, the second member **20** may be in direct contact with the surface of the second current collector **32**.

(18) Note that the bipolar battery of the present disclosure may include a form in which an intermediate member A is present between the first member **10** and the first current collector **31** and an intermediate member B (A and B may be identical members or members of the same type or may be different members) is present between the second member **20** and the second current collector **32**. When an intermediate member(s) is present on the first current collector **31** side and/or the second current collector **32** side in this manner, the cooling performance of the entire body including the intermediate members is compared between the first current collector **31** side and the second current collector **32** side. In other words, the cooling performance for the first current collector **31** exhibited by the first member **10** and the intermediate member A is higher than the cooling performance for the second current collector **32** exhibited by the second member **20** and the intermediate member B. Intermediate members may be present between the first member **10** and the first current collector **31** and between the second member **20** and the second current collector **32** as long as the cooling performance is higher on the first current collector **31** side than on the second current collector **32** side.

(19) As shown in FIG. 2, the second member **20** may be provided only on the other end side (a position facing the surface of the second current collector **32**) of the laminate electrode body **30** in the lamination direction. Alternatively, the second member **20** may have a portion which protrudes from the position facing the surface of the second current collector **32**, and, for example, the second member **20** may be present on the side surface side of the laminate electrode body **30** by bending the protruding portion.

(20) 1.3 Laminate Electrode Body

(21) The laminate electrode body **30** is arranged between the first member **10** and the second member **20**. The bipolar battery **100** can be charged and discharged by the battery reaction in the laminate electrode body **30**. As shown in FIG. 2, the laminate electrode body **30** comprises the first current collector **31** constituting a lamination direction end surface, the second current collector **32** constituting the other lamination direction end surface, the at least one bipolar current collector **33** arranged between the first current collector **31** and the second current collector **32**, and a plurality of power generating elements **34** which are electrically connected in series via the bipolar current collector **33** between the first current collector **31** and the second current collector **32**.

(22) 1.3.1 First Current Collector

(23) The first current collector **31** constitutes a lamination direction end surface of the laminate electrode body **30**, and is arranged between the first member **10** and the bipolar current collector **33**. In other words, the first current collector **31** is the outermost current collector arranged more outwardly on the first member **10** side than the bipolar current collector **33** of the laminate electrode body **30**.

(24) The first current collector **31** may be constructed from metal foil or metal mesh. From the viewpoint of high handleability, the first current collector **31** may be metal foil. The first current collector **31** may be composed of a plurality of layers of metal foil. Examples of the metal constituting the first current collector **31** may include Cu, Ni, Cr, Au, Pt, Ag, Al, Fe, Ti, Zn, Co, and stainless steel. Some sort of coating layer may be included on the surface of the first current collector **31** to adjust the resistance thereof. When the first current collector **31** is composed of a

plurality of layers of metal foil, some sort of layer may be included between the plurality of layers of metal foil. The thickness of the first current collector **31** is not particularly limited. It may be, for example, 0.1 μm or more or 1 μm or more, and may be 1 mm or less or 100 μm or less.

(25) The first current collector may be a positive electrode current collector or may be a negative electrode current collector. When the first current collector **31** is a positive electrode current collector, the second current collector **32**, which will be described later, is a negative electrode current collector, and when the first current collector **31** is a negative electrode current collector, the second current collector **32**, which will be described later, is a positive electrode current collector.

(26) 1.3.2 First Tab

(27) The first current collector **31** comprises a first tab **31a**. During charging of the bipolar battery **100**, electrical power is supplied to the power generating elements **34** via the first tab **31a**, and during discharging of the bipolar battery **100**, electrical power is discharged to the outside via the first tab **31a**. In other words, Joule heat generation due to energization can occur in the first tab **31a** during both charging and discharging of the bipolar battery **100**.

(28) The amount of heat generated by the first tab **31a** during energization of the battery **100** varies in accordance with the material and shape of the first tab **31a** (cross-sectional area, length, etc.), energization resistance with the members connected with the first tab **31a** for transferring current (for example, increased electrical resistance due to welding), and the Joule heat generation amount of the members connected to the first tab **31a**. The magnitude relationship between the amount of heat generated by the first tab **31a** and the amount of heat generated by the second tab **32a** can be confirmed by, for example, energizing the laminate electrode body **30**.

(29) The material of the first tab **31a** may be the same as the material of the first current collector **31** or may be different. The tab **31a** may be composed of a material having a high volume resistivity (electrical resistivity) as compared with that of the second tab **32a**. The thickness of the first tab **31a** may be the same as the thickness of the first current collector **31** or may be different. The cross-sectional area of the first tab **31a** may be smaller than the cross-sectional area of the second tab **32a**. It is sufficient that the first tab **31a** have a shape which protrudes from the first current collector **31**. As the protruding shape of the first tab **31a**, various shapes such as a polygonal shape, a semicircular shape, and a linear shape can be adopted. The method for providing the first tab **31a** on the first current collector **31** is not particularly limited. For example, the first tab **31a** may be formed by cutting out a part of the first current collector **31**, or may be formed by connecting the first tab **31a** to the first current collector **31** by welding or the like.

(30) The first tab **31a** may have a shutdown mechanism. The phrase “shutdown mechanism” means a mechanism for cutting off the current or making it difficult for current to flow when the first tab **31a** generates excessive heat, when excessive current flows through the first tab **31a**, or when the battery **100** is in an abnormal charging (overcharging) state. Specific examples of the shutdown mechanism include a heat-fusing mechanism, a mechanism using a PTC element, and a metal plate reversing mechanism by gas pressure. The phrase “heat-fusing mechanism” means that a notch (cut) is provided at the base of tab **31a**, and the notch (cut) is fused when the temperature rises or excessive current flows. The phrase “a mechanism using a PTC element” means that a PTC material (mixture of a resin and a conductive material; barium titanate; etc.) is arranged in a part of the current path of tab **31a**, and when the temperature rises, the resistance of the PTC material rises, making it difficult for current to flow. The phrase “metal plate reversing mechanism by gas pressure” means that when the internal pressure of the battery rises during overcharging and etc., the internal pressure inverts the metal plate, thereby cutting off the current (refer to, for example, Japanese Unexamined Patent Publication No. 2018-77977). When the first tab **31a** comprises a shutdown mechanism, the amount of heat generated during energization may be greater than when a shutdown mechanism is not provided.

(31) 1.3.2 Second Current Collector

(32) The second current collector **32** constitutes the other lamination direction end surface of the laminate electrode body **30**, and is arranged between the second member **20** and the bipolar current collector **33**. In other words, the second current collector **32** is the outermost current collector arranged more outwardly on the second member **20** side than the bipolar current collector **33** of the laminate electrode body **30**.

(33) The second current collector **32** may be constructed from metal foil or metal mesh. From the viewpoint of high handleability, the second current collector **32** may be metal foil. The second current collector **32** may be composed of a plurality of layers of metal foil. Examples of the metal constituting the first current collector **31** may include Cu, Ni, Cr, Au, Pt, Ag, Al, Fe, Ti, Zn, Co, and stainless steel. Some sort of coating layer may be included on the surface of the second current collector **32** to adjust the resistance thereof. When the second current collector **32** is composed of a plurality of layers of metal foil, some sort of layer may be included between the plurality of layers of metal foil. The thickness of the second current collector **32** is not particularly limited. It may be, for example, 0.1 μm or more or 1 μm or more, and may be 1 mm or less or 100 μm or less.

(34) 1.3.3 Second Tab

(35) The second current collector **32** comprises a second tab **32a**. During charging of the bipolar battery **100**, electrical power is supplied to the power generating elements **34** via the second tab **32**, and during discharging of the bipolar battery **100**, electrical power is discharged to the outside via the second tab **32a**. In other words, Joule heat generation due to energization occurs in the second tab **32a** during both charging and discharging of the bipolar battery **100**. However, the second tab **32a** has a small amount of heat generated during energization than the first tab **31a**. Thus, even if the second tab **32a** generates heat, a little or less heat diffuses into the second current collector **32**.

(36) The material of the second tab **32a** may be the same as the material of the second current collector **32** or may be different. The thickness of the second tab **32a** may be the same as the thickness of the second current collector **32** or may be different. It is sufficient that the second tab **32a** have a shape which protrudes from the second current collector **32**. As the protruding shape of the second tab **32a**, various shapes such as a polygonal shape, a semicircular shape, and a linear shape can be adopted. The method for providing the second tab **32a** on the second current collector **32** is not particularly limited. For example, the second tab **32a** may be formed by cutting out a part of the second current collector **32**, or may be formed by connecting the second tab **32a** to the second current collector **32** by welding or the like.

(37) The second tab **32a** may or may not comprise a shutdown mechanism. When the first tab **31a** comprises a shutdown mechanism, the second tab **32a** may not comprise a shutdown mechanism.

(38) 1.3.4 Bipolar Current Collector

(39) The bipolar current collector **33** is arranged between the first current collector **31** and the second current collector **32**. In FIG. 2, an aspect in which a plurality of bipolar current collectors **33** are provided in the laminate electrode body **30** is shown, and the number of bipolar current collectors **33** in the laminate electrode body **30** is not particularly limited as long as it is at least one. However, the case in which a plurality of bipolar current collectors **33** are provided is considered to highly exhibit the effect of the technology of the present disclosure.

(40) The bipolar current collector **33** may be constructed from metal foil or metal mesh. From the viewpoint of handleability, the bipolar current collector **33** may be metal foil. The bipolar current collector **33** may be composed of a plurality of layers of metal foil. Examples of the metal constituting the bipolar current collector **33** may include Cu, Ni, Cr, Au, Pt, Ag, Al, Fe, Ti, Zn, Co, and stainless steel. The metals constituting the first current collector **31**, the second current collector **32**, and the bipolar current collector **33** may be the same or may be different. Some sort of coating layer may be included on the surface of the bipolar current collector **33** to adjust the resistance thereof. Furthermore, when the bipolar current collector **33** is composed of a plurality of layers of metal foil, some sort of layer may be included between the plurality of layers of metal foil. The thickness of the bipolar current collector **33** is not particularly limited. For example, the

thickness may be 0.1 μm or more and may be 1 μm or more, and may be 1 mm or less or 100 μm or less.

(41) 1.3.5 Power Generating Elements

(42) It is sufficient that the power generating elements **34** be capable of causing a battery reaction to charge and discharge the bipolar battery **100**. For example, as shown in FIG. 2, each power generating element **34** may comprise a first active material layer **34a**, a second active material layer **34b**, and an electrolyte layer **34c** arranged between the first active material layer **34a** and the second active material layer **34b**. The plurality of power generating elements **34** are electrically connected in series via the bipolar current collector **33**.

(43) Among the first active material layer **34a** and the second active material layer **34b**, one is a positive electrode active material layer and the other is a negative electrode active material layer. When the first active material layer **34a** is a positive electrode active material layer, the first current collector **31** described above can become a positive electrode current collector, and when the first active material layer **34a** is a negative electrode active material layer, the current collector **31** described above can become a negative electrode current collector. Furthermore, when the second active material layer **34b** is a positive electrode active material layer, the second current collector **32** described above can become a positive electrode current collector, and when the second active material layer **34b** is a negative electrode active material layer, the second current collector **32** described above can become a negative electrode current collector.

(44) The positive electrode active material layer comprises at least a positive electrode active material. When the bipolar battery **100** is a solid-state battery, in addition to a positive electrode active material, a solid electrolyte, a binder, and a conductive agent can be optionally further included. Furthermore, when the bipolar battery **100** is a liquid electrolyte-based battery, in addition to a positive electrode active material, a binder and conductive agent can be optionally further included. A known active material may be used as the positive electrode active material. Between two materials having different potentials for occlusion and release (charging and discharging potentials) of a predetermined type of ions, the material demonstrating a noble potential can be used as the positive electrode active material and the material demonstrating a low potential can be used as a negative electrode active material, which is described later. For example, in the case of constituting a lithium ion battery, various lithium-containing composite oxides such as lithium cobalt oxide, lithium nickel oxide, $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$, lithium manganate, and spinel-based lithium compounds can be used as the positive electrode active material. When the bipolar battery **100** is a solid state battery, the surface of the positive electrode active material may be coated with an oxide layer such as a lithium niobate layer, a lithium titanate layer, or a lithium phosphate layer. When the bipolar battery **100** is a solid state battery, the solid electrolyte is preferably an inorganic solid electrolyte. This is because the ionic conductivity thereof is higher than that of organic polymer electrolytes. This is also because it has excellent heat resistance as compared with organic polymer electrolytes. Further, this is because it has excellent rigidity as compared with organic polymer electrolytes, whereby the bipolar battery **100** can be more easily constructed. Examples of preferred inorganic solid electrolytes may include oxide solid electrolytes such as lithium lanthanum zirconate, LiPON , $\text{Li}_{1-x}\text{Al}_x\text{Ge}_2\text{X}(\text{PO}_4)_3$, Li-SiO -based glasses, and Li-Al-S-O -based glasses; and sulfide solid electrolytes such as $\text{Li}_{2.5}\text{P}_{0.5}\text{S}_{5.5}$, $\text{Li}_{2.5}\text{SiS}_2$, $\text{LiI-Li}_{2.5}\text{SiS}_2$, $\text{LiI-Si}_{2.5}\text{P}_{0.5}\text{S}_{5.5}$, $\text{Li}_{2.5}\text{P}_{0.5}\text{S}_{5.5}\text{LiI-LiBr}$, $\text{LiI-Li}_{2.5}\text{P}_{0.5}\text{S}_{5.5}$, $\text{LiI-Li}_{2.5}\text{P}_{0.5}\text{O}_{5.5}$, $\text{LiI-Li}_{3.5}\text{PO}_4\text{P}_{0.5}\text{S}_{5.5}$, and $\text{Li}_{2.5}\text{P}_{0.5}\text{S}_{5.5}\text{GeS}_2$. In particular, sulfide solid electrolytes are preferred, and sulfide solid electrolytes comprising $\text{Li}_{2.5}\text{P}_{0.5}\text{S}_{5.5}$ are more preferable. Examples of the binder included in the positive electrode active material layer may include butadiene rubber (BR) based binders, butylene rubber (IIR) based binders, acrylate butadiene rubber (ABR) based binders, polyvinylidene fluoride (PVdF) based binders, and polytetrafluoroethylene (PTFE) based

binders. Examples of the conductive agent included in the positive electrode active material layer may include carbon materials such as acetylene black and Ketchen Black or metal materials such as nickel, aluminum, and stainless steel. The content of each component in the positive electrode active material layer may be the same as in the prior art. The shape of the positive electrode active material layer may also be the same as in the prior art. In particular, from the viewpoint of facilitating construction of the bipolar battery **100**, a sheet-like positive electrode active material layer is preferred. The thickness of the positive electrode active material layer is not particularly limited. For example, the thickness may be 0.1 μm to 2 mm. The lower limit thereof may be 1 μm or more and the upper limit thereof may be 1 mm or less.

(45) The negative electrode active material layer comprises at least a negative electrode active material. When the bipolar battery **100** is a solid state battery, in addition to the negative electrode active material layer, a solid electrolyte, binder, and conductive agent can further be optionally included. Furthermore, when the bipolar battery **100** is a liquid electrolyte-based battery, in addition to the negative electrode active material, a binder and a conductive agent can be optionally further included. A known active material may be used as the negative electrode active material. For example, when the battery is constituted as a lithium ion battery, a silicon-based active material such as Si, Si alloys, and silicon oxide; a carbon-based active material such as graphite or hard carbon; various oxide-based active materials such as lithium titanate; metallic lithium and lithium alloys can be used as the negative electrode active material. The solid electrolyte, the binder and the conductive agent can be appropriately selected and used from those exemplified for the positive electrode active material layer. The content of each component in the negative electrode active material layer may be the same as in the prior art. The shape of the negative electrode active material layer may also be the same as in the prior art. In particular, from the viewpoint of facilitating construction of the bipolar battery **100**, a sheet-like negative electrode active material layer is preferred. The thickness of the negative electrode active material layer is not particularly limited. For example, the thickness may be 0.1 μm to 2 mm. The lower limit thereof may be 1 μm or more and the upper limit thereof may be 1 mm or less.

(46) The electrolyte layer **34c** comprises at least an electrolyte. When the bipolar battery **100** is a solid state battery, the electrolyte layer **34c** can be a solid electrolyte layer comprising a solid electrolyte and optionally a binder. The solid electrolyte is preferably an inorganic solid electrolyte, as described above, and is particularly preferably a sulfide solid electrolyte. The binder can be appropriately selected and used from among those used for the positive electrode active material layer. The content of each component in the solid electrolyte layer may be the same as in the prior art. The shape of the solid electrolyte layer may also be the same as in the prior art. In particular, from the viewpoint of facilitating construction of the bipolar battery **100**, a sheet-like solid electrolyte layer is preferred. In this case, the thickness of the solid electrolyte layer may be, for example, 0.1 μm to 2 mm. The lower limit thereof may be 1 μm and the upper limit thereof may be 1 mm. Conversely, when the bipolar battery **100** is a liquid electrolyte-based battery, the electrolyte layer **34c** can comprise a liquid electrolyte and a separator. Known liquid electrolytes and separators may be used. Note that when comparing the case in which the electrolyte layer **34c** is a liquid electrolyte layer and the case in which it is a solid electrolyte layer, in the case in which the electrolyte layer **34c** is a solid electrolyte layer, construction of the bipolar battery **100** is considered to be easier.

(47) The power generating elements **34** and laminate electrode body **30** can be produced by known methods. The number of the plurality of power generating elements **34** included in the laminate electrode body is not particularly limited. For example, the number of power generating elements **34** may be 2 to 1000. The lower limit thereof may be 10 or more, and the upper limit may be 300 or less.

(48) Note that though the shape of the laminate surface (the surface shape of the first current collector **31**) of the laminate electrode body **30** is illustrated as rectangular in FIGS. **1** and **2**, the

shape of the laminate surface of the laminate electrode body **30** is not limited thereto.

(49) 1.3.6 Other Members

(50) In addition to the first member **10**, the second member **20**, and the laminate electrode body **30** described above, the bipolar battery **100** may comprise other members. For example, terminals connected to the tabs **31a**, **32a** may be provided. The terminals may be mechanically connected to tabs **31a** and **32a** so as to be detachable, or may be joined by welding.

(51) As described above, the bipolar battery **100** may comprise an exterior body **40** for accommodating the laminate electrode body **30**. In this case, the first member **10** may cool the first current collector **31** via the exterior body **40**. The exterior body **40** may be a laminate film or may be a metal case. However, even if an exterior body **40** is not provided, the bipolar battery can be constructed. For example, the side surfaces of the laminate electrode body **30** in the lamination direction may be sealed with a resin, and the sealing resin, the first current collector **31**, and the second current collector **32** may serve the same function as the extender body. In this case, the first member **10** may be in direct contact with the first current collector **31**, whereby the first current collector **31** can be directly cooled.

(52) As will be described later, the bipolar battery **100** may be connected to a heat exchanger **50**. As a result, the cooling efficiency of the first current collector **31** by the first member **10** is further improved.

(53) 2. Bipolar Battery Stack

(54) The technology of the present disclosure also includes an aspect as a bipolar battery stack. In other words, the bipolar battery stack of the present disclosure comprises a plurality of the bipolar batteries of the present disclosure described above. For example, the bipolar battery stack may be constructed by laminating a plurality of the bipolar batteries. The method for laminating the bipolar batteries is not particularly limited.

(55) 2.1 First Aspect

(56) FIGS. **3** and **4A** show a bipolar battery stack **1000** according to a first aspect. As shown in FIGS. **3** and **4A**, in the bipolar battery stack **1000**, the second member **20** of one bipolar battery **100** may be laminated on the first member **10** of another bipolar battery **100**. In other words, in the bipolar battery stack **1000**, one bipolar battery **100** and another bipolar battery **100**, which are adjacent, may be laminated in the same direction.

(57) 2.2 Second Aspect

(58) FIG. **4B** shows a bipolar battery stack **2000** according to a second aspect. As shown in FIG. **4B**, in the bipolar battery stack **2000**, the first member **10** of one bipolar battery **100** may be laminated on the first member **10** of another bipolar battery **100**. Furthermore, as shown in FIG. **4B**, in the bipolar battery stack **2000**, the second member **20** of one bipolar battery **100** may be laminated on the second member **20** of another bipolar battery **100**. In other words, in the bipolar battery stack **2000**, one bipolar battery **100** and another bipolar battery **100**, which are adjacent, may be laminated in mutually opposite directions.

(59) 2.3 Third Aspect

(60) FIG. **4C** shows a bipolar battery stack **3000** according to a third aspect. As shown in FIG. **4C**, in the bipolar battery stack **3000**, a first member **10** may be shared by one bipolar battery **100** and another bipolar battery **100**. Furthermore, as shown in FIG. **4C**, in the bipolar battery stack **3000**, a second member **20** may be shared by one bipolar battery **100** and another bipolar battery **100**. In other words, in the bipolar battery stack **3000**, first members **10** and second members **20** may be shared between adjacent bipolar batteries **100** while omitting some first members **10** and some second members **20**.

(61) 2.4 Other Aspects

(62) As in the bipolar battery stack **4000** shown in FIG. **5**, the first member **10** may be connected to a heat exchanger **50**. For example, when the first member **10** has coolant passages, the coolant passages may be connected to the heat exchanger **50**, whereby coolant may be circulated between

the heat exchanger **50** and the first member **10**. The location where the heat exchanger is connected in the bipolar battery stack is not particularly limited. Among the side surfaces (the surfaces other than the laminated surfaces) of the bipolar battery **100**, when the surface on which tabs **31a**, **32a** protrude is the first side surface, the surface opposite to the first side surface is the second side surface, and the two side surfaces between the first side surface and the second side surface are the third side surface and the fourth side surface, the heat exchanger **50** may be connected to the second side surface of the bipolar battery **100**, or the heat exchanger **50** may be connected to the third side surface or fourth side surface.

(63) The bipolar battery stack may comprise arbitrary members other than those described above. For example, wiring or a stack case may be provided.

(64) 3. Effects

(65) In conventional bipolar batteries, during energization of the battery, the tabs connected to the outermost current collectors generate heat, and this heat may diffuse from the tabs toward the interior of the electrode body via the outermost current collectors. The amount of heat generated during energization of the battery differs between the positive electrode side tab and the negative electrode side tab provided in the bipolar battery. For example, when one tab is composed of Al and the other tab is composed of Cu, and the shapes of both tabs are the same, since the electrical resistivity of Al is greater than that of Cu and the thermal conductivity of Al is less than that of Cu, the tab composed of Al has a greater amount of heat generated during energization and has a lower heat dissipation capacity than the tab made of Cu. Thus, when comparing the current collector connected to Al tab and the current collector connected to Cu tab, a significant amount of heat is conducted and diffused to the former current collector, and thermal deterioration of the electrode body is likely to occur in the vicinity of the former current collector. In connection thereto, according to the bipolar battery of the present disclosure, the outermost current collector (first current collector) connected to the tab (first tab) having a significant amount of amount of heat generated during energization can be appropriately cooled by the first member, which has high cooling performance. As a result, thermal deterioration of the laminate electrode body in the vicinity of the first current collector connected to the first tab, which generates significant amounts of heat, can be suppressed.

INDUSTRIAL APPLICABILITY

(66) The bipolar battery and bipolar battery stack of the present disclosure is widely applicable, from small power sources such as those for mobile devices to large power sources, such as those for electric vehicles.

REFERENCE SIGNS LIST

(67) **10** first member **20** second member **30** laminate electrode body **31** first current collector **31a** first tab **32** second current collector **32a** second tab **33** bipolar current collector **34** power generating element **34a** first active material layer **34b** second active material layer **34c** electrolyte layer **40** exterior body **50** heat exchanger **100** bipolar battery **1000**, **2000**, **3000**, **4000** bipolar battery stack

Claims

1. A vehicle comprising a bipolar battery, the bipolar battery comprising a first member, a second member, and a laminate electrode body arranged between the first member and the second member, wherein the laminate electrode body comprises a first current collector constituting a lamination direction end surface, a second current collector constituting the other lamination direction end surface, at least one bipolar current collector arranged between the first current collector and the second current collector, and a plurality of power generating elements which are electrically connected in series via the bipolar current collector between the first current collector and the second current collector, the first current collector is arranged between the first member and the

bipolar current collector, the second current collector is arranged between the second member and the bipolar current collector, the first current collector has a first tab, the second current collector has a second tab, an amount of heat generated by the first tab during energization of the battery is greater than an amount of heat generated by the second tab, the first member is a cooling member for cooling the first current collector, and a cooling performance of the first member is greater than a cooling performance of the second member.

2. The vehicle according to claim 1, wherein the first tab has a shutdown mechanism.

3. The vehicle according to claim 1, wherein the second member is a heat insulating member.

4. The vehicle according to claim 1, wherein the vehicle comprises a bipolar battery stack, the bipolar battery stack comprises a plurality of the bipolar batteries.

5. The vehicle according to claim 4, wherein the second member of one of the bipolar batteries is laminated on the first member of another bipolar battery.

6. The vehicle according to claim 4, wherein the first member of one of the bipolar batteries is laminated on the first member of another bipolar battery.

7. The vehicle according to claim 4, wherein the first member of one of the bipolar batteries is shared by another of the bipolar batteries.
